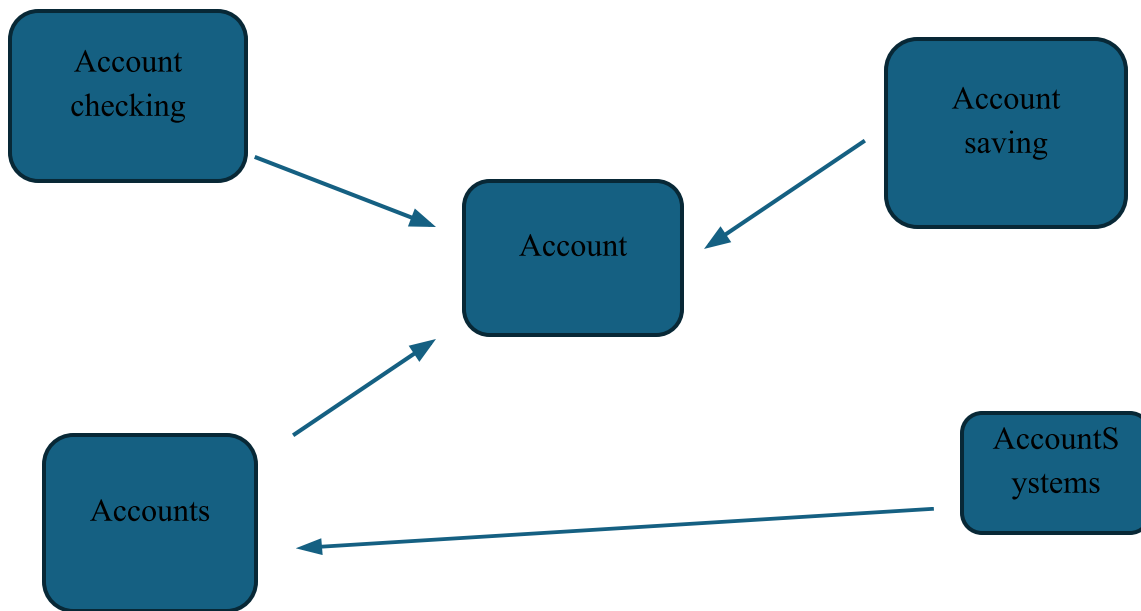




Account

Attributes (private)

- int* accountID
- string customerName
- float balance
- int accountType // 1 = checking, 2 = saving

Methods (public)

- Account()
- virtual ~Account()
- Account(const Account& other)
- Account& operator=(const Account& other)

Mutators:

- void SetAccountID(int id)
- void SetCustomerName(const string& name)
- void SetBalance(float amount)
- void SetAccountType(int type)

Accessors:

- int GetAccountID() const
- string GetCustomerName() const
- float GetBalance() const
- int GetAccountType() const

Operations:

- virtual void deposit(float amount)
- virtual bool withdraw(float amount)
- virtual Account& operator+=(Account& src)

AccountChecking : public Account

Attributes (private)

- float transactionFee

Methods (public)

- AccountChecking()
- AccountChecking(int id, const string& name, float initialBalance, float fee)
- AccountChecking(const AccountChecking& other)
- AccountChecking& operator=(const AccountChecking& other)
- virtual ~AccountChecking()

Mutators / accessors:

- void SetTransactionFee(float fee)
- float GetTransactionFee() const

Overrides:

- virtual void deposit(float amount) override
- virtual bool withdraw(float amount) override

AccountSaving : public Account

Attributes (private)

- float minimumBalance

Methods (public)

- AccountSaving()
- AccountSaving(int id, const string& name, float initialBalance, float minBal)
- AccountSaving(const AccountSaving& other)
- AccountSaving& operator=(const AccountSaving& other)
- virtual ~AccountSaving()

Mutators / accessors:

- void SetMinimumBalance(float minBal)
- float GetMinimumBalance() const

Overrides:

- virtual bool withdraw(float amount) override

Accounts

Attributes (private)

- vector<Account*> data // no more static array / maxsize
- csce1040_hw3_fa25

Methods (public)

- Accounts()
- ~Accounts()

Operations:

- bool add(Account* acct)
- int find(int accountId) const // index or -1
- Account* at(int i) const
- int size() const

AccountSystem

Attributes (private)

- Accounts accts

Methods (public)

- AccountSystem()

User operations / I/O:

- void addAccount()
- void deposit()
- void withdraw()
- void transfer()
- void printAll() const

Pseudocode

Add Account (`AccountSystem::addAccount`)

Add Account
Prompt user for account ID
Check if ID already exists in Accounts collection
If ID exists
Print error message
Return
Prompt user for customer name
Prompt user for initial balance
Prompt user for account type (1 = Checking, 2 = Saving)
If type is Checking
Prompt user for transaction fee
Create AccountChecking object
Set ID, name, balance, transaction fee
Add object pointer to Accounts collection
If type is Saving
Prompt user for minimum balance
Create AccountSaving object
Set ID, name, balance, minimum balance
Add object pointer to Accounts collection

Deposit (`AccountSystem::deposit`)

Deposit
Prompt user for account ID
Find account in Accounts collection
If account not found
Print error message
Return
Prompt user for deposit amount
Call account's deposit method with amount

Withdraw (`AccountSystem::withdraw`)

Withdraw
Prompt user for account ID
Find account in Accounts collection
If account not found
Print error message
Return
Prompt user for withdrawal amount
Call account's withdraw method with amount
If withdraw returns false

Print error message
Else
Print success message

Transfer (AccountSystem::transfer using operator+=)

Transfer
Prompt user for source account ID
Prompt user for destination account ID
Find source account in Accounts collection
Find destination account in Accounts collection
If either account not found
Print error message
Return
If source and destination are the same account
Print error message
Return
Call destinationAccount operator+= with sourceAccount

Transfer Logic (Account::operator+=)

Transfer (operator+=)
Prompt user for transfer amount
If amount is less than or equal to zero
Print error message
Return this account
Call withdraw on source account with amount
If withdraw returns false
Print transfer failed message
Return this account
Call deposit on this account with amount
Print transfer successful message
Return this account

Base Deposit (Account::deposit)

Deposit (base Account)
If amount is less than or equal to zero
Return
Add amount to balance

Base Withdraw (Account::withdraw)

Withdraw (base Account)
If amount is less than or equal to zero
Return false

If amount is greater than balance
Return false
Subtract amount from balance
Return true

Checking Deposit (AccountChecking::deposit)

Deposit (checking)
Compute netAmount as amount minus transactionFee
If netAmount is less than or equal to zero
Print error message
Return
Call base deposit with netAmount

Checking Withdraw (AccountChecking::withdraw)

Withdraw (checking)
Compute total as amount plus transactionFee
Call base withdraw with total
Return result

Saving Withdraw (AccountSaving::withdraw)

Withdraw (saving)
Get current balance
If current balance minus amount is less than minimumBalance
Print error message
Return false
Call base withdraw with amount
Return result

Add to Collection (Accounts::add)

Add Account to collection
If pointer is null
Return false
Get ID from account
Search collection for matching ID
If a match is found
Print error message
Return false
Add account pointer to vector
Return true

Find in Collection (Accounts::find)

Find Account by ID
Loop through vector of account pointers
For each account
If account ID matches requested ID
Return index
If no match found
Return -1

Print All Accounts (`Accounts::PrintAllAccounts`)

Print All Accounts
If vector is empty
Print message that there are no accounts
Return
For each account in vector
Print account ID, customer name, balance, and account type